Energy Price Prediction using Transformer Architecture

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Abstract. OpenRemote is a player in the asset management game, with a focus on energy-related assets. For its costumers, a fine-grained knowledge of energy prices would open the doors for price optimisation. OpenRemote's current solution, that uses the Prophet model, is able to capture Min and Max levels with relative accuracy, but falls short on the daily variations in the baseline, due to its deterministic model architecture. This research aims to test multiple neural network architectures in search for one that can beat the Prophet model. Our findings show that the iTransformer architecture, along with other Transformer models, perform best at Energy Price Prediction.

1 Introduction

Energy Price Forecasting is a highly sought after technique that adds value to companies that benefit from load balancing and management. With the further decarbonisation of the energy grid, renewable energy introduces a strong temporal characteristic to the challenge; one that Neural Networks, particularly advancements in RNN[1, 23, 18, 5, 13], Transformer[10, 21, 15, 22] and ensemble[2, 4, 7] technology can leverage to improve the accuracy of predictions. As feature extraction mechanisms grow more robust[4, 1, 23, 16, 18], and models involve complex sets of features like weather[17, 21], location, and energy consumption[16], the demand for better, more accurace, and more farreaching models is growing withing the industry.

One of the industry players with a keen interest in the technology is OpenRemote. The company provides a platform where entities with multiple energy-demanding components can manage their resources. One of the microservices that OpenRemote provides is an Energy Price Forecasting (EPF) model hosted by Prophet, that does not employ Neural Networks for its predictions[20]. Their current service is only able to offer 24h predictions ahead of time, which causes constraint for schedueled load balancing, and limits the amount of saving a client could make. OpenRemote argues that a 48h prediction window would be more benefitial. Furthemore, the Prophet model is more accurate around the peaks, but relatively innacurate anywhere else. This is due to the limitations of the modeling techniques employed which, although easy to configure and faster than a Neural Network to train and generate forecastings, struggles with the complexities of multi-feature temporal and spacial data that RNNs and Transformers capture with greater ease.

This paper will present, out of a collection of methods collected via a thourough literature review, the results of experiments conducted with the intention of comparing different combinations of methods - both preprocesing, and neural network architectures. The end result if the analysis and thourough review of the results. It will also provide advice on how to implement the techniqeus reviewed throughut this research.

2 Methods

The objective of this research is to interrogate a variety of methods within the literature surrounding time-series regression. More specifically, energy price or energy related time-series prediction. This particular research field adopts a multi-feature, and often multimodal approach to the problem. This section outlines the methods used for the research, in hopes of finding a model architecture that outperforms the current method that OpenRemote employes.

2.1 Research Design

The research is designed to almost perform as a census on the available methodologies. A healthy variety of methods for preprocessing and architectures were choosen and divided into experiments that compile similar architectures, with sub experiemnts which introduce small variations. These variations intend to reveal the effect of a particular method i.e. if the difference between experiment 1.a and 1.b is one specific architecture, then the difference in performance might have to do with the introduction of that architecture.

The scope of the research is strictly for time-series energy price prediction, and its goal of being more accurate than the benchmark.

We will, for the purposes of this proposal, use a naming convention of Exp.$i.v$, where i is the number of the experiment configuration, and v is a letter, whose order is reset for each i.

2.1.1 Exp.1

From our literature, one of the most recurrent topics is the power of ensemble models, particularly ones using RNNs. To start out simple, we will introduce a CNN-LSTM variant architecture as a base and design multiple experiments that build upon it.

Since we are using CNNs, which serve the purpose of capturing long-term sequential patters, it would be ideal to test an array of preprocessing methods, some of which enhance 3-dimentional features. For Exp.1, all variations will cycle through:

1. Normalisation
2. Wavelet Transform[16, 6]
3. Transformer Encoding (PatchTST[14])

Exp.1.a Variation **a** uses a CNN-BiLSTM architecture as the baseline recurrent model for Exp.1. The CNN frontend extracts local temporal patterns from the multivariate sequence and compresses them into a representation that is easier for the sequence model to exploit. The BiLSTM then models dependencies in both temporal directions, which is a common strategy in energy forecasting and other multivariate time-series tasks[18, 5]. This variant therefore acts as the reference configuration against which the newer recurrent and attention-based variants can be compared.

Exp.1.b Variation **b** replaces the BiLSTM block with an xLSTM module. xLSTM extends the classical LSTM design with updated memory and gating mechanisms intended to preserve long-range information more effectively while remaining recurrent in nature[3]. Although it was introduced in a different domain, more recent work shows that xLSTM-style models can be transferred successfully to multivariate time-series forecasting[8, 11]. This variation is included to test whether a more modern recurrent architecture can improve upon the conventional BiLSTM baseline without abandoning the recurrent paradigm entirely.

Exp.1.c Variation **c** uses a CNN-BiLSTM-Transformer architecture. In this configuration, the CNN and BiLSTM primarily serve as hierarchical feature extractors, while the Transformer encoder refines the interactions among the resulting sequence representations. The purpose of this variant is to test whether attention complements recurrent modelling when both are placed in the same pipeline, as hybrid CNN-LSTM-Transformer systems have reported strong forecasting performance in related energy applications[2, 23]. Because this model combines convolution, recurrence, and attention in a single architecture, the effect of each preprocessing method should be examined carefully for this variation.

Exp.1.d Variation **d** uses a CNN-Transformer architecture and removes the recurrent stage used in Exp.1.a–c. This creates a direct comparison between hybrid recurrent-attention models and a purely convolutional-attention pipeline, where the CNN captures local temporal structure and the Transformer models broader dependencies across the encoded sequence. The purpose of this variation is to test the null hypothesis that, once attention is introduced, the recurrent block no longer contributes useful information and may instead introduce redundancy or noise. Exp.1.d therefore serves as the cleanest control for determining whether the gains seen in Exp.1.c come from the Transformer itself or from the interaction between recurrence and attention.

This set of variations offers a broad comparison between conventional recurrent forecasting models, newer recurrent alternatives, and hybrid attention-based architectures, while keeping the preprocessing strategies fixed across the experiment. In that sense, Exp.1 provides the bridge between the more classical neural approaches in the literature and the Transformer-focused models that are explored more directly in Exp.2[2, 23, 13].

	Exp.1			
	a	b	c	d
Architecture	CNN-BiLSTM	CNN-xLSTM	CNN-BiLSTM-Transformer	CNN-Transformer
Preprocessing methods	Norm., WT, PatchTST	Norm., WT, PatchTST	Norm., WT, PatchTST	Norm., WT, PatchTST
Additional notes				To prove null-hypothesis i.e. RNNs punish accuracy

Table 1: Exp.1: Model Architectures and Preprocessing

	Exp.2				
	a	b	c	d	e
Architecture	Decoder-Only Transformer	Decoder-Only + FFT	Encoder-Decoder Transformer	Kernel-based Transformer	iTransformer
Preprocessing methods	Standard	Wavelet Transform	Standard	Kernel-based	Standard
Additional notes		Same model as **a**			

Table 2: Exp.2: Model Architectures and Preprocessing

2.1.2 Exp.2

Current trends in AI strongly favour Transformer architectures, or Attention layers. Exp.2 will focus on exploring this particular aspect in more detail. Exp.1 also features Transformers, but in this section, we want to amplify their strenghts by means of expanding our number of features. Non-linearity is accutely endemic to all of our features[13]. One hypothesis is that the Transformer architecture will be able to capture linear features where there is usually non-linearity.

Exp.2.a Variation **a** will use a classic multi-head Decoder-Only transformer[15]. Decoder-Only transformers perform well in renewable scenario analysis, and use less memory[7]. This is relevant for our study because the nature of the data and analysis is similar by virtue of the non-linear and complex time-series tasks both try to address.

Exp.2.b Variation **b** will use the same model as Exp.2.a, but utilise Wavelet Transformers for feature extraction[24].

Exp.2.c Variation **c** will use a multi-head Encoder-Decoder Transformer. Encoder-Decoder variants perform great with forecasting tasks[7].

Exp.2.d Variation **d** will use a Kernel-based Transformer. This architecture can better capture linearity when it exists[19] which has been pointed out as a reason for transformers failing to generalise on Time-Series tasks[9] , all the while using less memory.

Exp.2.e Variation **e** will use an inverted Transformer, henceforth known as iTransformer; a method introduced by a landmark paper back in 2023[12]. The paper specifically demonstrates the effectiveness of the proposed methodology for Time-Series Forecasting, and has been explicitely used for EPF[21].

2.2 Data

The dataset comprehends a meryad of features from different sources, such as metereological data, overall energy consumption and production, and gas prices. The target variable, being the energy price, is also included. The $\vec{\Delta}$ between samples is 1 hour.

The data is also preprocessed using the methods above. Timestamp data is feature engineered to decompose it into many features, such as hour, day of week, month, quarter and weekend, all with cyclical scaling.

2.3 Procedure

All experiment models are based on the Torch library. For the preprocessing, there is a mixture of self-written methods, and some from scikit-learn and pywt, for Wavelet transform. They are arranged into the experiment configurations described above, and run sequentially using a custom script, that outputs the results in bulk.

2.4 Preprocessors

The preprocessors used in this experiment merit a section of their own, as they are an integral part of the research, and overlwhelm the project with a veneer of complexity alike the ensemble models described above. We will go through the main preprocessing methods, and explain what they constitute; provide a logical explanation for the methodology employed. Some preprocessors might discriminate the experiment that they belong to.

2.4.1 Standard Preprocessor (EXP. 2)

The standard processor applies simple processing on the data. For floating points, it applies a standard scaler, and a MinMax scaler. For datetime data, it applies Cyclical Scaling.

Standard Scaler The standard scaler simply uses Standard Normal Distribution to normalise the values. This helps with uneven scales within the dataset. So the mean of each column becomes 0 and the standard deviation becomes 1. The MinMaxScaler scales the data in a max to min scale. So the minimum value is 0 (or -1 if there are negative values for that feature) and the max value becomes 1.

Both are used because, while standard scaling is useful for baseline detection, the MinMax Scaler ensures that we dont lose outlier feedback.

Cyclical Scaling This scaling technique is used for date and time, and its purpose is to enforce the circular dimention of time onto the data, using trigonometry, as presented in 1. This is one of the valid methods for processing date and time columns for time-series data. This particular method provides cyclicality to time. Like how months, weekdays and hours are not linear but cyclical; they 'reset' on every cycle. This method hightlights that property. Thusly, we extract more features out of the original date.

2.4.2 Discrete Wavelet Transform Preprocessor (Exp.2.b)

A Discrete Wavelet Transform preprocessor is used for decomposing the signal into more nieche frequencies. This technique allows for nitpicking which 'bands' or 'signals' we want to remove, either because they capture a pattern we want to obscure, or are just noise that the model can do nothing with.

Initially, we wrote down in the proposal that we would use Fast Fourier Transform, but for time-series data, the wavelet transform is more useful. The amplitude of the signal is not as useful as the decomposition of said signal, since it will allow us to reduce noise for some features, while preserving the overall patterns.

Consider Figure 2, where we can see in its totality, the temperature data in our dataset. Notice how there are minor imperfections in an otherwise clear periodical fluctuation. That is to be expected, as the Netherlands follows a pretty typical temperature cicle.

Now take a look at Figure 3. The signal from earlier is now decomposed into 3 different signals, called bands. Band 1 (or d1) looks like a lot of noise, same as Band 2. However, Band 3 seems a little more constrained. Altough it clearly detects peaks, it also finds stability. That is what we are looking for when doing Wavelet analysis.

Furthermore, by performing ablation[1], we are able to get a clearer picture of the effects of each band to a potential model.

[1] Ablation is the process of recurrently testing the same model on the same dataset, with the only difference being the removal of a given feature, for every feature in the dataset.

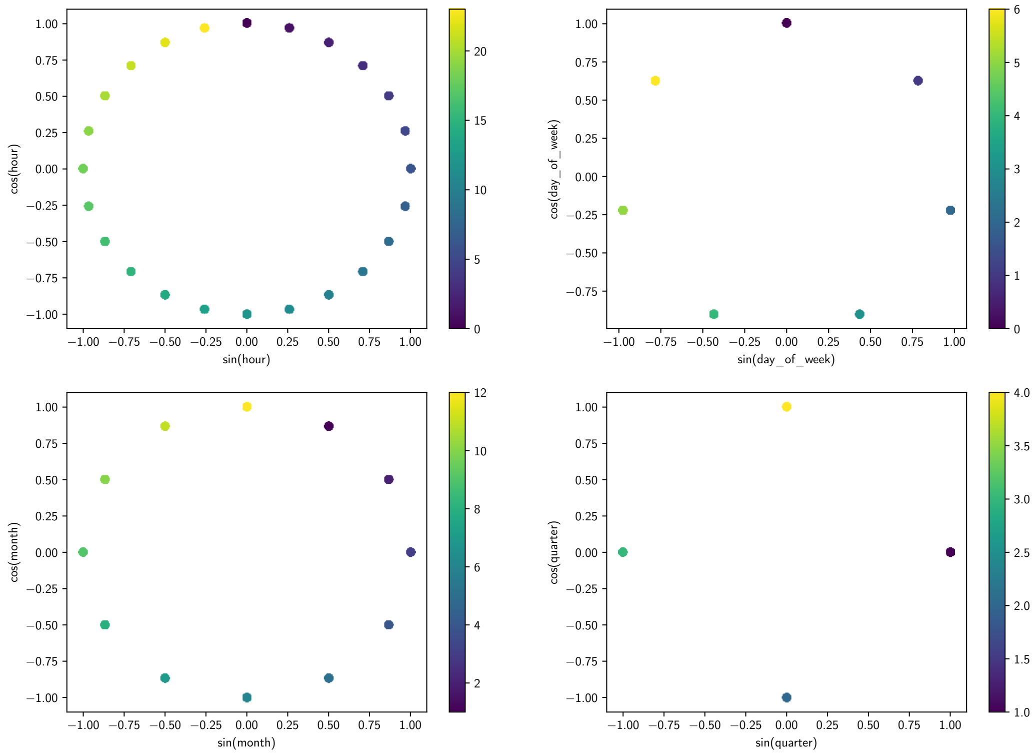

Figure 1: A plot of date features after being subjected to Cyclical Scalling

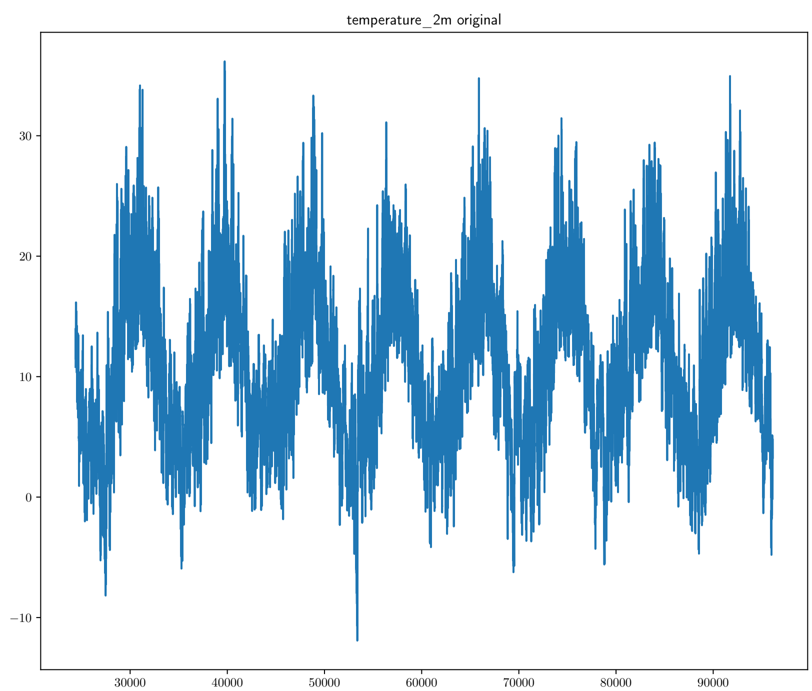

Figure 2: A plot of the temperature over time in our dataset

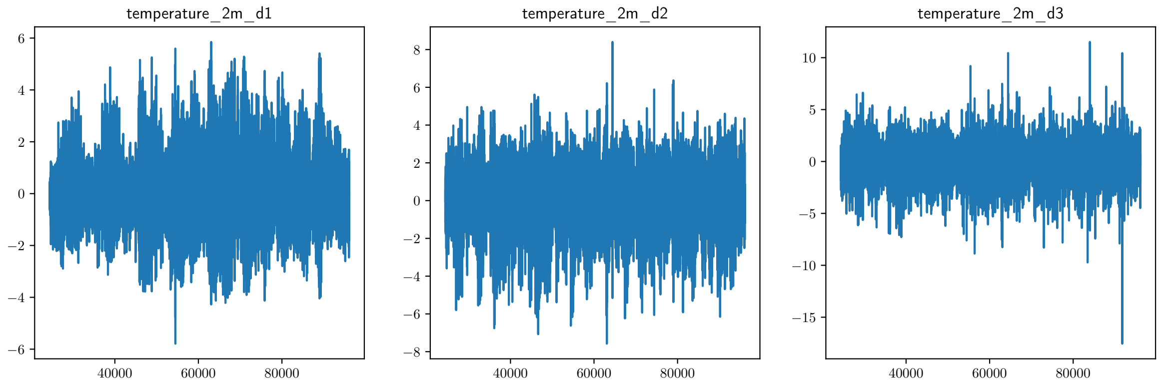

Figure 3: A plot of the temperature over time in our dataset after Wavelet decomposition

variable	band	rmse	delta
total_load	d2	33.756189	0.249319
generation_forecast	d1	33.642795	0.135925
price	d1	33.632644	0.125774
temperature_2m	d3	33.614782	0.107912
Low	d2	33.605844	0.098975
wind_speed_10m	d3	33.603512	0.096642
Price	d3	33.599684	0.092814
price	d2	33.589829	0.082959
cloud_cover	d2	33.589608	0.082739
total_load	d3	33.579247	0.072377
temperature_2m	d2	33.574455	0.067585
shortwave_radiation	d2	33.569995	0.063125
High	d2	33.561575	0.054705
total_load	d1	33.552758	0.045888
Change %	d2	33.546079	0.039209
shortwave_radiation	d1	33.543296	0.036426
generation_forecast	d3	33.535409	0.028539
Low	d3	33.533773	0.026903
Change %	d3	33.532598	0.025728
cloud_cover	d3	33.520752	0.013882
wind_speed_10m	d2	33.519621	0.012751
High	d3	33.517491	0.010621
High	d1	33.515562	0.008692
Low	d1	33.513244	0.006374
Price	d1	33.508143	0.001273
cloud_cover	d1	33.501218	-0.005652
temperature_2m	d1	33.500676	-0.006193
shortwave_radiation	d3	33.496145	-0.010725
generation_forecast	d2	33.475815	-0.031055
Price	d2	33.473631	-0.033239
price	d3	33.470569	-0.036301
wind_speed_10m	d1	33.422056	-0.084814
Change %	d1	33.381141	-0.125729

Table 3: A table that shows the results of the ablation test per band and feature individually

We can read table 2.4.2 by simply looking at the column delta and understanding that the bigger the number is, the more benefitial it is to remove the associated band for that feature. If the number is negative, it means that the band is benefitial. Its worth mentioning that this method has limitations, as the base algorithm a Light Gradient Boosting Machine is a very simple model in contrast with our architectures. Therefore, the effects of removing a given band on this model might differ dramatically for the other models.

2.4.3 Kernel-based Preprocessing (Exp.2.d)

Exp.2.d pairs the Kernel-based Transformer with kernel-embedded feature mapping. Rather than an explicit standalone preprocessor, the kernel mechanism within the model architecture implicitly projects inputs into a higher-dimensional feature space, allowing the model to better capture linear structure in otherwise non-linear time-series signals.

3 Results

For our performance metrics, we will assume RMSE and MAE. The difference between the two, for the purposes of this paper, is their tolerance to outlier data. MAE is more representative of the typical performance, while RMSE is outlier sensitive. So we could interpret a better MAE as more adapt for normal circumstances. Conversely, we can interpret RMSE as a good metric for how well the models can handle outlier data.
Furthermore, all the experiment were run on 100 epochs.
The results determine that Experiment 2 and Experiment 1.d outperform the Prophet model, as

Table 4: Table with the results of the experiments, including the Prophet model

run_title	duration	mae	rmse
Prophet baseline	03:44:00	35.705599	51.297253
Mean baseline	00:00:01	41.667505	55.593261
Zero baseline	00:00:00	87.795577	100.832967
MinMax + Mean	00:00:02	41.667505	55.593261
Exp1.a CNN-BiLSTM + Norm	00:01:27	39.738697	54.504961
Exp1.b CNN-xLSTM + Norm	01:40:47	36.616713	51.924452
Exp1.c CNN-BiLSTM-Tx + Norm	00:02:41	39.024715	52.396946
Exp1.d CNN-Tx + Norm	00:01:55	**35.202300**	**49.788010**
Exp1.a CNN-BiLSTM + Wavelet	00:02:03	38.347490	51.884933
Exp1.b CNN-xLSTM + Wavelet	02:44:48	37.643475	52.044102
Exp1.c CNN-BiLSTM-Tx + Wavelet	00:04:25	37.002768	**50.413288**
Exp1.d CNN-Tx + Wavelet	00:02:31	36.766987	51.588176
Exp1.a CNN-BiLSTM + Patch	00:00:55	42.927602	54.618952
Exp1.b CNN-xLSTM + Patch	00:11:38	43.010114	55.008874
Exp1.c CNN-BiLSTM-Tx + Patch	00:01:44	40.832303	54.211177
Exp1.d CNN-Tx + Patch	00:01:26	42.898806	54.757634
Exp2.a Decoder-Only Tx + Exp2 Standard	00:02:52	**31.805239**	**45.796124**
Exp2.b Decoder-Only Tx + Exp2 Wavelet	00:06:11	36.966030	**50.014923**
Exp2.c Encoder-Decoder Tx + Exp2 Standard	00:03:43	**30.480106**	**44.597015**
Exp2.d Kernel Tx + Kernel	00:03:32	**33.739584**	**48.833119**
Exp2.e iTransformer + Exp2 Standard	00:07:32	**30.210997**	**42.766888**

shown in Table 4, with the results above the baseline highlighted in bold. The improvementes range from 1.7% to 16.6% for RMSE and 1.4% to 15.3% for MAE. However, most experiments failed to go past the threshold, especially those who did not feature a transformer model. It is also worth pointing out that Exp1.b and Exp1.c only differ in their inclusion of the RNN component (it being the BiLSTM). The experiment without it performed better, beating the baseline.
There is also an observable difference in the performance of the preprocessors. The preprocessor that performs the best is the Standard preprocessor for Exp2, which averages to 44.3 RMSE, followed by Wavelet with 51.1 averafe RMSE. This difference of 6.8% is statistically significant ($p = 0.014$). The Kernel variant also performed rather well, tho it only has 1 datapoint.

4 Discussion and Advice

Our results paint a very clear picture transformer models outperform classic time-series neural networks, even when ensembled. We can posit that the inclusion of CNN or an RNN increase the complexity of the model without any benefits. This experiment inadvertidly functioned as a replication study of Yong Liu et. al. paper on iTransformers[12], whose conclusions on the effectiveness of the inverted transformer are replicated in our experiment, inspite the different application. Another paper on inverted transfomers applies the method on a very different context, but not only does it reach the same conclusions, but they benchmark their results against an encoder-only transformer, and a model which includes an LSTM[25]. The iTransformer performs the best, followed by the encoder-noly transformer, a PatchTST, and the Seq2Seq LSTM+Attention.
These papers results being analogous to ours matter because they validate our methods by means of replication. However, the question of whether an ensemble model with the iTransformer would improve its ability to adapt to certain conditions, like predicting outliers, or the Min and Max of the data. A path for future research would look into what the stakeholder would like to optimise for and pick a model architecture that excels in that. The preprocessors are another point of improvement, particularly for Wavelet decomposition.

One of the limitations of this group is the level of expertise and comfortableness with data science concepts, and their proper utilisation. This limitation results in the application of methods with methodological flaws. That the wavelet decomposition performed significantly worse than the standard could be due to poor feature engineering. It also completely isolated temporal features, which are properly augmented in the standard preprocessor, and likely contributed to its sucess. Future research should focus on improving feature selection based on ablation, and introducing better treatment of temporal features, with the hypothesis being that the best of both preprocessors could bring about better performance.

With that said, the results from the standard preprocessor corroborate the idea that temporal features need to be strongly emphasised, along with the proper scalling of numerical features.

Another hypothesis left unanswered is the effect of CNNs on transformers, since the transformer in Exp1 is different from the ones in Exp2. We can't event posit that the LSTM component is not benefitial to accuracy, since the difference between those including an LSTM and those without (in Exp1) is not statistically significant ($p = 0.3$). Even if we just look at the experiment were the only difference between the models is the inclusion of the RNN, the difference is still not significant ($p = 0.4$). However, none of the experiments retires the CNN. So that hypothesis is still up for answering. Future work could consider an ensenbmle of the iTransformer and CNN.

5 Conclusion

OpenRemote is looking for a better model that can predict energy prices more accurately, with emphasis on the small daily variations. This research aims to test multiple neural network techniques and bruteforce a model that would beat OpenRemote's current implementation, which is the Prophet model, utilising neural networks, which produce models with a less deterministic profile. Multiple experiments were set up, containing deviations in between them, architeture-wise and in the preprocessors used. Experiment 1 looks into classic neural networks, proved to work for time-series data, and Experiment 2 focuses on Transformer models. The results favour significantly the Transformer models, which boast improvements upwards of 16.6% and 15.3% for RMSE and MAE respectively. The best model architecture uses an iTransformer, which achieved an RMSE of 42.7 and a MAE of 30.2. Improvements in feature engineering and the preprocessor techniques could improve model performance in future works.

6 Limitations

This work is limited by the size of the group, and the experience with the field of research explored in this project, being the field of AI. The quality of this work is impacted by this factor, and it should not be downplayed. Time has also been a limiting factor. Different team members had differing availabilities, but the nature of the semester also limits the capacity for actual work. An example of this is the time that it takes to document, validate, and upkeep our portfolios.

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